Two-Dimensional Magnetotelluric Modeling Based on Stochastic Path Integral: TE Case

Hongyu Zhou, *Graduate Student Member, IEEE*, Maokun Li, *Fellow, IEEE*, Fan Yang, *Fellow, IEEE*, and Shenheng Xu, *Member, IEEE*

***Abstract*—In this study, we investigate the application of the stochastic path integral (SPI) for 2-D magnetotelluric (MT) modeling. Our goal is to assess whether electromagnetic (EM) numerical algorithms can mechanistically adapt to parallel heterogeneous computing architectures and maximize the conversion of computational power into efficient EM field simulations. Using the Feynman–Kac formula, we derive a path integral representation of the MT Helmholtz equation in a stochastic framework. We can then investigate the online–offline two-stage MT-SPI algorithm that includes Monte Carlo (MC) random walks, mapping matrix construction, and large-scale matrix–vector multiplication. Numerical experiments verify the correctness and stability of the proposed SPI algorithm. At the relative error of 1.2%, SPI simultaneously achieves over 550× acceleration and approximately 30% memory reduction on GPU under a MATLAB-based implementation, compared with the finite difference method (FDM).**

***Index Terms*—Computational electromagnetics, magnetotellurics (MTs), Monte Carlo (MC) simulation, numerical modeling, parallel computing, random walk, stochastic algorithm.**

I. Introduction

ELECTROMAGNETIC (EM) modeling provides a ubiquitous tool for EM field analyses by efficiently and accurately solving Maxwell's equations. It is widely used in engineering and research. For instance, Magnetotelluric (MT) modeling computes the Earth's response to the incident natural EM fields. Accurate and efficient MT modeling is essential for stable inversion and reliable interpretation of MT data [1].

Traditional MT forward modeling methods mainly include the finite difference method (FDM) [2], integral equation method [3], and finite element method (FEM) [4]. These methods involve converting Maxwell's equations into a linear matrix equation, which is then solved with direct or iterative solvers [5], [6], [7]. However, when the number of unknowns is large, solving the linear system can be computationally intensive and time-consuming. Many approaches have been explored to accelerate MT modeling, including the Rayleigh-FFT technique [8], local mesh approach [9], and deep learning methods [10], [11].

Parallel computing is an effective approach to solve large-scale EM field modeling and other. In parallel computing, large numbers of unknowns are distributed across multiple computing nodes and solved separately, with boundary continuity ensured through synchronization and communication between processors. Over the past decades, regional decomposition methods [12], [13], [14], high efficiency assembly of basis function matrix [15], [16], [17], and efficient iterative solution algorithms for large-scale sparse matrix equations [18], [19], [20] have been extensively studied. However, EM algorithms inherently require frequent communication between computing units. Therefore, as the problem scale grows, load balancing and I/O bottlenecks will present significant challenges.

Recently, stochastic approaches have gained attention in computational EMs [21], [22]. Stochastic algorithms decompose the solution of the EM field equations into a large number of independent subtasks, making them inherently parallel with minimal communication requirements and high scalability. The Feynman–Kac formula extends the deterministic Feynman integral into a probabilistic framework, offering a probabilistic functional solution to deterministic partial differential equations (PDEs) through the path integral of stochastic processes [23], [24]. By representing the EM field solution with numerous independent random walks, this method naturally demonstrates both parallelism and scalability [25]. Chati et al. [26] discussed the possibility of using stochastic path integral (SPI) directly to solve the Helmholtz equation. Janaswamy [27] adopted SPI to model the paraxial wave propagation in homogeneous media. Budaev and Bogy [28] transformed the Helmholtz equation into both the Eikonal equation and the complete transport equation and solved the latter with the Feynman–Kac formula. A recent paper [22] on 1-D MT modeling based on SPI demonstrated its capability to model inhomogeneous media with complex boundary conditions (BCs). However, SPI has not yet been applied to 2-D EM modeling involving complex BCs and inhomogeneous media.

In this work, we study 2-D MT modeling based on SPI. The solution of the Helmholtz equation is derived based on the Feynman–Kac formula, and solving the matrix equation is converted to Monte Carlo (MC) random walk and matrix–vector multiplication. Numerical experiments verify the accuracy and stability of the proposed method and show the potential of the proposed method to accelerate computation with parallel heterogeneous computing resources.

Received 16 May 2025; revised 5 August 2025 and 18 September 2025; accepted 2 October 2025. Date of publication 6 October 2025; date of current version 27 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 624B2085, in part by the National Key Research and Development Program of China under Grant 2023YFB3905003, in part by the Science and Technology Project of China National Petroleum Corporation (CNPC) under Grant 08-01-2024, and in part by the Fundamental Research Funds for the Central Universities under Grant ZYGX2025TS007 of UESTC. *(Corresponding author: Maokun Li.)*

The authors are with the Department of Electronic Engineering, Beijing National Research Center for Information Science and Technology (BNRist), Tsinghua University, Beijing 100084, China (e-mail: maokunli@tsinghua.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2025.3618304, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2025.3618304

II. Method

A. 2-D MT Modeling (TE Mode)

With $e^{j\omega t}$ notation, the electric field Helmholtz equation for MT modeling can be derived from Maxwell's equations [2]

$$\boldsymbol{\nabla}^2\mathbf{E} + k^2\mathbf{E} = 0 \tag{1}$$

where k is the wavenumber

$$k = \sqrt{\omega^2\mu\left(\epsilon - j\frac{\sigma}{\omega}\right)}. \tag{2}$$

Here, ω is the angular frequency, μ is the permeability, ϵ is the permittivity, and σ is the conductivity. In the 2-D problem, we solve the x-component of the electric field (E_x) within a rectangular computational domain G in the yoz plane. Neglecting the permittivity of the conductive Earth as a low-frequency approximation, and assuming thatthe permeability μ to be constant, Helmholtz equation (1) for 2-D transverse electric (TE) mode MT inside the computational domain G can be expressed with

$$\frac{\partial^2 E_x}{\partial y^2} + \frac{\partial^2 E_x}{\partial z^2} - j\omega\mu\sigma E_x = 0. \tag{3}$$

A feasible BC setting is to attribute Dirichlet BC to the top and Neumann BC to the sides of the computational domain, that is,

$$E_x|_{\text{top}} = \text{Constant}, \quad \left.\frac{\partial E_x}{\partial y}\right|_{\text{sides}} = 0. \tag{4}$$

Assuming that the conductivity is constant below the lower boundary, the electric and magnetic fields at the lower boundary satisfy

$$\frac{E_x}{H_y} = \sqrt{\frac{\mu}{-j\sigma/\omega}}. \tag{5}$$

Substituting H_y with the vertical derivative of E_x, we have the Robin BC for the lower boundary

$$\left.\frac{\partial E_x}{\partial z} + \sqrt{j\omega\mu\sigma}E_x\right|_{\text{lower}} = 0. \tag{6}$$

In MT modeling, we compute the apparent resistivity and phase on the surface of the survey. For the TE mode, they are defined as follows:

$$\rho_{\text{TE}} = \frac{1}{\omega\mu}\left|\frac{E_x^0}{H_y^0}\right|^2, \quad \phi_{\text{TE}} = \arctan\left(\frac{E_x^0}{H_y^0}\right). \tag{7}$$

The above equations can be summarized as a boundary value problem (BVP)

$$\left.\frac{1}{2}\nabla^2 E_x + BE_x\right|_{\text{int(G)}} = 0 \tag{8}$$

$$\left.a\nabla E_x + bE_x + f\right|_{\partial G} = 0 \tag{9}$$

where int(G) and ∂G represent the interior and boundary of the computational domain G, respectively. B, a, b, and f are formulated with

$$B = -\frac{1}{2}j\omega\mu\sigma \tag{10}$$

$$(a, b, f) = \begin{cases} (0, 1, C_0), & \text{at } \partial G_{\text{D}} \\ (1, 0, 0), & \text{at } \partial G_{\text{N}} \\ \left(1, \sqrt{j\omega\mu\sigma}, 0\right), & \text{at } \partial G_{\text{R}} \end{cases} \tag{11}$$

where ∂G_{D}, ∂G_{N}, and ∂G_{R} denote the Dirichlet, Neumann, and Robin boundaries, respectively, and C_0 is a constant related to the strength of the natural EM field source. Generally speaking, (8) and (9) are solved after transformed into a linear equation system, using some numerical algorithms like FDM or FEM.

B. Stochastic Representation of MT Modeling

In addition to conventional methods, (8) and (9) can be solved with a stochastic approach. One possible framework is based on the Feynman–Kac formula, which converts the time-domain PDEs into the expectation of a stochastic differential equation [24]. With further mathematical derivation, this approach can also be applied to compute the EM field in the frequency domain [27].

In the discrete $\mathbb{R}^2$ space, we define a discrete-time Markov chain $\{\boldsymbol{\Xi}_n = (\Xi_{1,n}, \Xi_{2,n}),\ n \in \mathbb{N}\}$ associated with (8) and (9) (as shown in [24], [26], [27], and [28]). We further define $d\boldsymbol{\Xi}_n = \boldsymbol{\Xi}_{n+1} - \boldsymbol{\Xi}_n$, and

$$P(d\boldsymbol{\Xi}_n) = \begin{cases} \frac{1}{4}, & \boldsymbol{\Xi}_n \in \text{int(G)} \wedge d\boldsymbol{\Xi}_n = \{(0, \pm 1) \vee (\pm 1, 0)\} \\ 1, & \{\boldsymbol{\Xi}_n \in \partial G_{\text{D}} \wedge d\boldsymbol{\Xi}_n = (0, 0)\} \text{ or} \\ & \{[\boldsymbol{\Xi}_n \in \partial G_{\text{N}} \vee \boldsymbol{\Xi}_n \in \partial G_{\text{R}}] \wedge d\boldsymbol{\Xi}_n = -a\mathbf{n}\} \\ 0, & \text{else.} \end{cases} \tag{12}$$

Here, a is defined in (11), and $\mathbf{n}$ is the normal vector at the boundary. $\boldsymbol{\Xi}_n$ represents the coordinates of a virtual particle performing a *random walk* within the region G. The particle moves with equal probability to one of the four adjacent locations when inside the domain G. At the Neumann or Robin boundary, it is reflected back into the interior along the negative normal direction, while at the Dirichlet boundary, the walk terminates.

Denoting the initial location of the random walk with $\boldsymbol{\Xi}_0(\boldsymbol{\Xi}_0 \in G)$, we express the x-component electric field E_x at $\boldsymbol{\Xi}_0$ by the expectation of a path integral function $\Phi_{\boldsymbol{\Xi}_0}(\cdot)$

$$E_{x,\boldsymbol{\Xi}_0} = \int_\Omega \Phi_{\boldsymbol{\Xi}_0}\left(\boldsymbol{\xi}_n\right) P\left(d\boldsymbol{\xi}_n\right) \tag{13}$$

where

$$\Phi_{\boldsymbol{\Xi}_0}\left(\boldsymbol{\xi}_n\right) = f\left(\boldsymbol{\xi}_\tau\right) e^{\int_0^\tau B(\boldsymbol{\xi}_s)ds + b(\boldsymbol{\xi}_s)d\lambda_s}. \tag{14}$$

Here, Ω denotes the sample space, $P(d\boldsymbol{\xi}_n)$ represents the probability measure on the sample space, and $\boldsymbol{\xi}_n$ is a realization of the stochastic process. f, B, and b are the functions of the coordinates, as defined in (8)–(11). τ is the time when the particle first reaches the Dirichlet boundary. ds and $d\lambda_s$ represent the infinitesimal time intervals during the entire random walk or when the particle reaches the Neumann or Robin boundary, respectively. Since the expectation expression in (13) is difficult to compute analytically, we approximate it by calculating the average of N_{path} randomly sampled probability functionals $\Phi_{\boldsymbol{\Xi}_0}(\boldsymbol{\xi}_{n,i})$, which are realized through MC simulation. These sampled probability functionals $\Phi_{\boldsymbol{\Xi}_0}(\boldsymbol{\xi}_{n,i})$ are computed using (14), taking into account the entire random walk trajectory.

Replacing B and b with the expressions introduced in (10) and (11) and expressing the integral using matrix–vector multiplication, we can further rewrite (13) and (14) as follows:

$$E_{x,\boldsymbol{\Xi}_0} = \frac{1}{N_{\text{path}}}\mathbf{I}_{1\times N_{\text{path}}}\boldsymbol{\Phi}_{\boldsymbol{\Xi}_0} \tag{15}$$

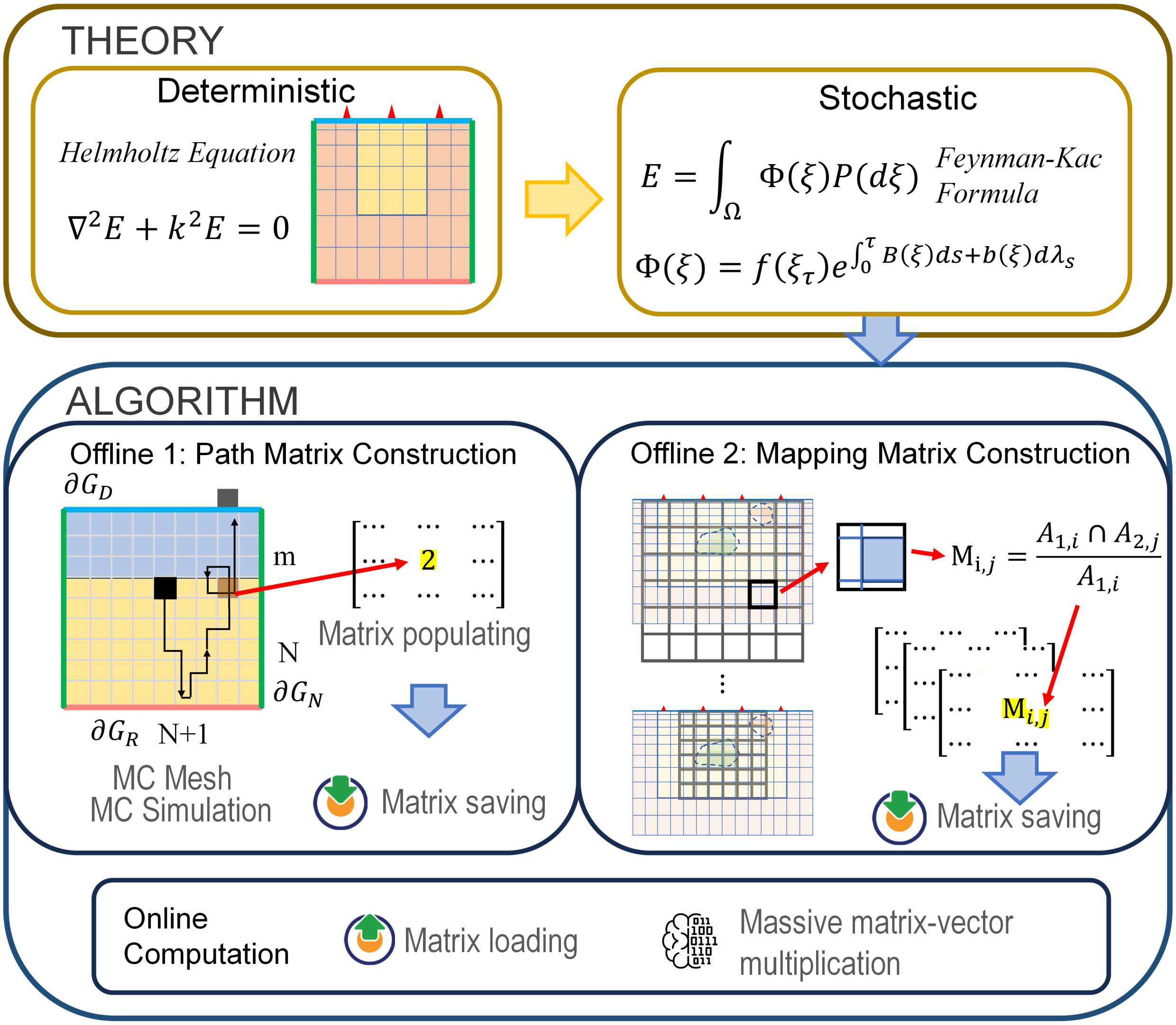

Fig. 1. Schematic of MT-SPI. The algorithm consists of two offline stages and one online computational stage.

where

$$\mathbf{\Phi}_{\Xi_0} = \exp\left\{-\frac{1}{2} j\omega\mu L^2 \mathbf{P}_{\Xi_0}\boldsymbol{\sigma} - \left(j\omega\mu L^2\right)^{\frac{1}{2}} \mathbf{P}_{\Xi_0,\mathrm{R}} \left(\boldsymbol{\sigma}_\mathrm{R}\right)^{\frac{1}{2}}\right\}. \quad (16)$$

Here, ω is the angular frequency of the EM field, and L is the unit length of the model grid, which is uniform in this study. $\mathbf{I}_{1\times N_{\mathrm{path}}}$ is a row vector of ones. $\boldsymbol{\sigma}$ and $\boldsymbol{\sigma}_\mathrm{R}$ are the conductivity vectors of the entire model and the Robin boundary, respectively. $\mathbf{P}_{\Xi_0}$ and $\mathbf{P}_{\Xi_0,\mathrm{R}}$ are, respectively, the total path matrix and the Robin boundary path matrix, corresponding to all N_{path} random walk realizations starting from Ξ_0. The shape of $\mathbf{P}_{\Xi_0}$ is $N_{\mathrm{path}} \times N_z N_x$, where the $(i,\ j)$th element represents the number of times the ith random walk path visits the jth pixel in the mesh. $\mathbf{P}_{\Xi_0,\mathrm{R}}$ is a subset of $\mathbf{P}_{\Xi_0}$, with the shape being $N_{\mathrm{path}} \times N_x$, where $(i,\ j)$th element denotes the number of times the ith random walk path visits the jth pixel on the Robin boundary. N_z and N_x are the number of pixels in the vertical and horizontal directions of the mesh. In the MT section shown in Fig. 1, the Dirichlet, Neumann, and Robin BCs are, respectively, assigned to the top, sides, and bottom boundaries.

C. MT-SPI Algorithm With Detached Phases

Equations (15) and (16) formulate the fundamental theory of SPI-based MT modeling. However, the computation for multiple receivers at multiple frequencies, which is typically required in MT modeling, still needs to be addressed. One straightforward approach is to repeatedly conduct an MC simulation with particles starting from multiple observation points and directly assign multiple angular frequency values to ω in (16). However, this approach may lead to several issues. First, as will be shown, each MC simulation may require hundreds of thousands of individual paths, making the computation expensive. Second, we find from numerical experiments that it is challenging to achieve high computational accuracy for all frequencies with a single discretized grid.

To address these issues, we point out that the MT-SPI simulation can be divided into an offline step to construct and store precalculated matrices, as well as an online step to conduct matrix–vector multiplication on demand. The considerations encompass, first, the EM field measured at a specific frequency and location is primarily influenced by the nearby conductivity distribution, as described by the *footprint* theory (the locally sensitive region of the EM wave resembles its "footprint") [29], [30], [31]. Second, as shown in (15) and (16), the path matrix is independent of the conductivity distribution, suggesting the possibility of decoupling the setup of the path matrices and constructing conductivity vectors that can each be operated by a single path matrix. We, therefore, construct a three-stage MT-SPI algorithm, and the schematic is shown in Fig. 1. First, a separate MC mesh apart from the model mesh is designed. On the former mesh, the path matrix is simulated and saved, while on the latter mesh, the subsurface conductivity model is defined. Next, we design an interpolation strategy that allows different areas in the survey region to be mapped onto a single MC mesh based on the frequency and receiver location. Finally, we compute the product of the path matrix and a series of conductivity vectors. The three stages are elaborated in detail as follows.

1) Path Matrix Construction: The shape of the square MC mesh is $(N+m)\times(N+1)$. The conductivity of the upper m rows of pixels is fixed to 1×10^{-8}, modeling the air, while the conductivity of the lower N rows vary, modeling the subsurface medium. The value of m can be heuristically determined; here, we set it to 5. Dirichlet, Neumann, and Robin BCs are assigned to the top, sides, and bottom of the mesh, respectively. A large number of virtual particles are released at the $(m+1, N/2+1)$ pixel, which is named the *release point*, and meander within the MC mesh according to (12). At the mean time, the path matrix $\mathbf{P}_{\Xi_0}$ is populated. Once all random walk realizations are terminated, the path matrix is completed and stored in memory. This procedure can be conducted offline.

2) Mapping Matrix Construction: For each frequency and receiver, a distinct mapping matrix is constructed depending on the physical location of the MC mesh, including the unit physical length depending on the frequency, and the location depending on the receiver location.

The unit physical lengths l of the MC mesh can be calculated with

$$l(f)=\frac{503\alpha}{N}\sqrt{\frac{\rho}{f}} \tag{17}$$

with ρ being the subsurface resistivity in average, f being the frequency, and α being a hyperparameter to be configured before computation. Equation (17) implies that the MC mesh is extended to a depth of α times the skin depth [32]. Due to the tradeoff between accurately representing the conductivity distribution and capturing the attenuation depth of the EM wave, the computational accuracy does not vary monotonically with α. The optimal value of α can be determined by minimizing the relative computing error between SPI and the reference responses, which will be mentioned in Section III-A. The location of the MC mesh is determined by aligning the *release point* with the receiver location in the survey region. In this way, the physical region corresponding to the MC mesh for any given frequency and receiver is fully determined.

Adopting the linear averaging-over-size (AoS) scheme [33], we construct $\mathbf{M}_{f,\mathbf{r}}$, a set of sparse matrices with shape $(N+1)N\times N_zN_x$, to map the model mesh onto the lower N rows of the MC mesh. The (i, j)th element in $\mathbf{M}_{f,\mathbf{r}}$ represent the weight of the overlap between the jth model mesh pixel $\mathcal{A}_{2,j}$ and the ith MC mesh pixel $\mathcal{A}_{1,i}$ area, defined as follows:

$$\mathbf{M}_{f,\mathbf{r}}(i,j)=\frac{\mathcal{A}_{1,i}\cap\mathcal{A}_{2,j}}{\mathcal{A}_{1,i}} \tag{18}$$

where $\cap$ denotes the overlapping. This procedure can also be conducted offline.

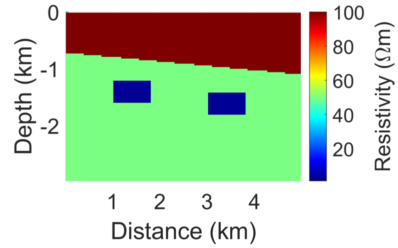

Fig. 2. Synthetic model.

3) Online Matrix–Vector Multiplication: The online computation of MT response relies on the path matrix $\mathbf{P}_{\Xi_0}$ and the mapping matrices $\mathbf{M}_{f,\mathbf{r}}$s generated during the offline stage. These matrices are first loaded into RAM or GPU memory. Given any conductivity model $\boldsymbol{\sigma}$ defined on the model mesh, the conductivity on the MC mesh is computed by

$$\mathbf{v}(f,\mathbf{r})=\mathbf{M}_{f,\mathbf{r}}\boldsymbol{\sigma} \tag{19}$$

where $\mathbf{r}$ denotes the receiver location. There are in total $N_\mathrm{f}N_\mathrm{r}$ different $\mathbf{v}$ vectors. Next, (20) and (21) are used to compute the electric field at receiver $\mathbf{r}$, which are slight modifications of (15) and (16)

$$E_{x,(f,\mathbf{r})}=\frac{1}{N_\mathrm{path}}\mathbf{I}_{1\times N_\mathrm{path}}\boldsymbol{\Phi}_\mathbf{r} \tag{20}$$

and

$$\boldsymbol{\Phi}_{f,\mathbf{r}}=\exp\left\{-\frac{1}{2}j\omega\mu l^2\mathbf{P}_{\Xi_0}\mathbf{v}-\left(j\omega\mu l^2\right)^{\frac{1}{2}}\mathbf{P}_{\Xi_0,\mathrm{R}}\left(\mathbf{v}_\mathrm{R}\right)^{\frac{1}{2}}\right\}. \tag{21}$$

The electric fields one pixel above or below the receiver are computed with the same approach, with which the magnetic field and then apparent resistivity and phase can be computed using (7).

III. Numerical Experiments

A. Preparations

We first introduce the setup of the computational domain, definition of the performance metrics, and configuration of parameters and hyperparameters in both the FDM and SPI solvers. The size of the computational domain is 5 km in width and 3 km in depth. The 12 receivers are evenly distributed along the Earth's surface, and 13 working frequencies of 0.001, 0.003, 0.01, 0.032, 0.1, 0.316, 1, 3.16, 10, 31.6, 100, 316, and 1000 Hz are employed. We generate a synthetic model with two layers and two blocks, as shown in Fig. 2. Dark red, grass green, and dark blue represent resistivities of 100, 50, and 3 $\Omega\cdot\mathrm{m}$, respectively. We also generate a test model set, consisting of 2000 models with much more complex conductivity distribution: the number of blocks ranges from 1 to 20, and the resistivities of both the background and the blocks are independently sampled from the uniform

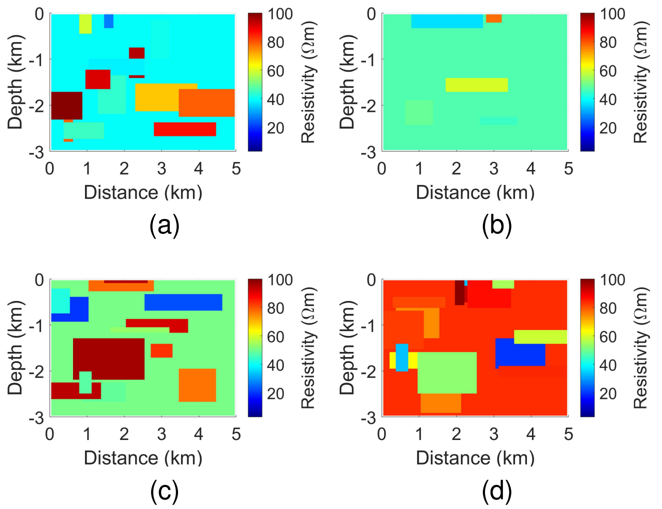

Fig. 3. (a)–(d) Visualizes four models selected from the test model set of 2000 cases.

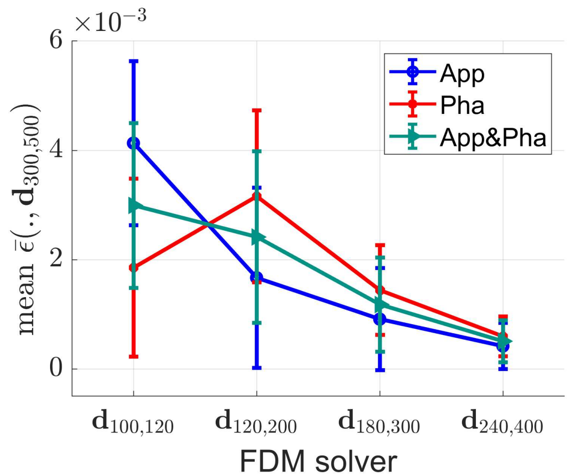

Fig. 4. Means and STDs of MREs between different FDM solvers. App and Pha, respectively, represent the apparent resistivity and phase.

distribution $\mathcal{U}(20, 100)\ \Omega\cdot$m. Four example models from the test set are displayed in Fig. 3.

Denoting two separately computed MT responses using different solvers with $\mathbf{d}_1$ and $\mathbf{d}_2$, the relative error image (REI, $\boldsymbol{\epsilon}$) can be computed with

$$\boldsymbol{\epsilon}(\mathbf{d}_1, \mathbf{d}_2) = \left|\frac{\mathbf{d}_1 - \mathbf{d}_2}{\mathbf{d}_2}\right|. \tag{22}$$

The division in (22) is performed elementwise, so $\mathbf{d}_1$, $\mathbf{d}_2$, and $\boldsymbol{\epsilon}$ all share the same shape of $nN_\mathrm{r} \times N_\mathrm{f}$. Here, n is either 1 (for apparent resistivity or phase section individually) or 2 (for the full data section). N_r and N_f, respectively, represent the number of MT receivers and working frequencies. The mean relative error (MRE, $\bar{\epsilon}$) can, then, be expressed as follows:

$$\bar{\epsilon} = \frac{1}{nN_\mathrm{r}N_\mathrm{f}}\Sigma_{i=1}^{nN_\mathrm{r}}\Sigma_{j=1}^{N_\mathrm{f}}\boldsymbol{\epsilon}_{i,j}. \tag{23}$$

Throughout this article, we use MRE and REI to quantify the relative data discrepancy between different solvers.

In this work, the MT response computed with FDM is used as the reference for comparison. The FDM solver adopted in this work is consistent with those in [34] and [35] and is briefly introduced in the Appendix. To make sure the FDM result is accurate enough, five different discretizations are tested, respectively, being 120 × 200, 180 × 300, 240 × 400, and 300 × 500, uniform in both horizontal and vertical directions. Using the five FDM solvers, data responses of all the 2000 cases in the test model set are computed, and the means and standard deviations (STDs) of the MREs between the FDM with 300 × 500 discretization [abbreviated with FDM (300, 500)] and other four solvers are calculated and displayed in Fig. 4. As the grid size increases, the mean MRE reduces from 3×10^{-3} to 5×10^{-4} monotonically, and the std narrows down from 1.5×10^{-3} to 3.9×10^{-4}, clearly demonstrating the convergence of the adopted FDM solver. Therefore, throughout this work, the true value of the MT response is approximated with the data computed by the FDM (300, 500) solver.

The optimal value of the hyperparameter α is determined by minimizing the mean of MRE between SPI and FDM (300, 500) over a set of models $\mathcal{D}$, that is,

$$\hat{\alpha} = \arg\min_{\alpha} \frac{1}{|\mathcal{D}|}\Sigma_{k=1}^{|\mathcal{D}|}\bar{\epsilon}_k \tag{24}$$

in which $|\mathcal{D}|$ represent the size of $\mathcal{D}$. In this work, $|\mathcal{D}|$ is constructed with the four models shown in Fig. 3. We densely sampled α in the range of 2–5, and finally, set it to 3.33.

B. Experiment 1

The proposed MT-SPI method is validated on the model shown in Fig. 2. The 16 different path matrix configurations are tested, where the combinations of (N, N_path) are selected from $\{30, 40, 60, 80\}$ and $\{0.1, 0.3, 0.5, 1\}$M. As introduced in Section II-C, $N = 30$ represents that the SPI mesh has 35 grids in the vertical direction and 31 grids in the horizontal direction, and the other cases including $N = 40, 60$, and 80 are handled similarly. For each configuration, five independent path matrices are sampled by MC simulation. Therefore, there are in total 80 independently sampled path matrices, which are, respectively, used to compute both the MT responses of the model in Fig. 2 and then the corresponding REIs and MREs. Fig. 5 demonstrates three scatter plots for the 80 MRE data points. Specifically, we focus on four representative parameter combinations: (N, N_path) = (30, 0.1M), (40, 0.3M), (60, 0.5M), and (80, 1M). For each of the four parameter settings, one out of the five SPI responses is selected and visualized in Fig. 6, as well as the corresponding REIs. As N and N_path increase, the mean MRE over different path matrix realizations decreases, and the std of MRE across different realizations narrows, confirming that the accuracy and stability of MT-SPI improve with increasing simulation parameters. Due to space limitations, more results are included in the supplementary materials. Figs. S1 and S2 visualize the apparent resistivity, phase, and REIs out of the five SPI responses for each of the 16 SPI parameter combinations. Figs. S3–S5 visualize the SPI responses computed by all five path matrix samples for selected SPI parameter combinations, as well as the corresponding REIs.

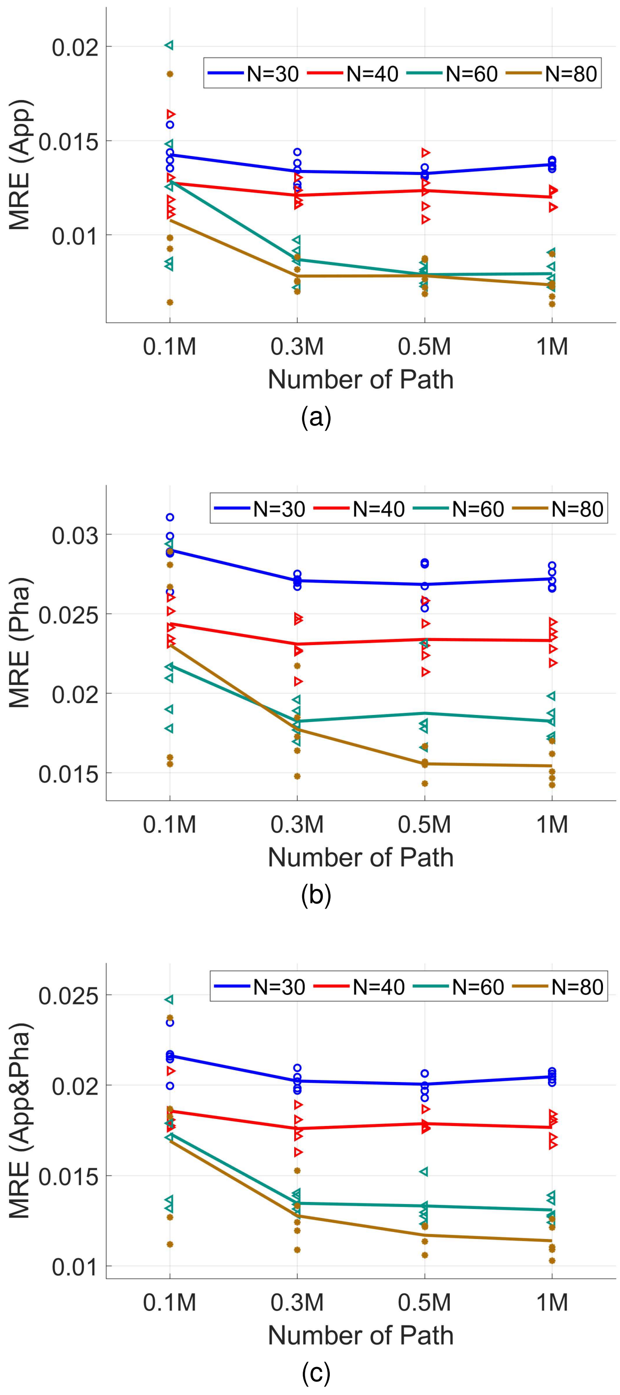

Fig. 5. Mean relative error distributions of (a) apparent resistivity (App), (b) phase (Pha), and (c) full data (App&Pha) sections computed by five independently realized path matrices in each of the SPI simulation settings.

We report some interesting observations from the aforementioned figures, include the following.

1) The average MRE decreases monotonically with the mesh size N, and with increasing N_{path} it first decreases then stabilizes.
2) The variance of MRE yielded by independently sampled path matrices decreases with the path number N_{path} and increases with the mesh size N.

We analyze that the relative error $\bar{\epsilon}$ can be approximately divided into a deterministic part $\bar{\epsilon}_{\text{d}}$ and a random part $\bar{\epsilon}_{\text{r}}$. Among the two parameters, N primarily influences $\bar{\epsilon}_{\text{d}}$ through the *discretization error* [36]. As N increases, the discretization error decreases, leading to a monotonic reduction in the overall relative error. Mainly influenced by N_{path}, $\bar{\epsilon}_{\text{r}}$ can be seen as the randomized *convergence error* [36], related to the ratio between the number of samples and the size of the probability space. With an increasing N_{path}, the sampling of the probability space becomes more thorough; therefore, the integral defined in (13) can be better estimated. However, reducing the random convergence error does not affect the deterministic error, so the overall relative error will not decrease indefinitely with N_{path}. On the other hand, as N increases, the probability space enlarges. Therefore, a fixed number of sampled paths appears less sufficient, hence emphasizing more local behavior and leading to less stable estimations.

C. Experiment 2

We respectively compute the MT sections of the 2000 models using FDM (300, 500) and SPI with the same (N, N_{path}) combinations as in Section III-B. As the SPI parameters increase from (30, 0.1M) to (60, 0.5M), the mean MREs for apparent resistivity, phase, and full data decrease by 66%, 30%, and 51%, respectively, clearly demonstrating the convergence of the SPI algorithm with increasing parameter size. The mean MRE for SPI (80, 1M) is slightly higher than SPI (60, 0.5M), which is mainly due to the randomness in the path sampling and the path matrix realization. Given that the difference in mean MRE between these two configurations is relatively small, this anomaly does not impact our overall assessment of the effectiveness of the SPI method. For each of the four parameter configurations, the MREs of all 2000 models are distributed within a relatively compact range. MT sections for the four models shown in Fig. 3 computed with SPI (80, 1M), as well as the REIs, are, respectively, visualized in Figs. 7 and 8. The histogram counts of the 2000 MREs of the apparent resistivity, phase, and full data sections are respectively shown in Fig. 9(a)–(c), and their statistical summaries are provided in Table I. Please refer to Fig. S6 in the supplementary materials for more visualized models and corresponding REIs.

As a comparison, FDM solvers with nonuniform meshes, that is, N_{Fx} represent the horizontal grids with uniform sizes, and N_{Fz} represent the vertical grids with the thickness forming a geometric sequence with the first term of d_{F} meters and a common ratio of k_{F}, are also tested. We use three parameter combinations, i.e., (N_{Fz}, N_{Fx}, d_{F}, k_{F}) = (25, 30, 19.3, 1.13), (50, 60, 10.3, 1.06), and (100, 120, 4.93, 1.03), respectively, and the means and STDs of MRE are also summarized in Table I. With the parameter scale increasing, the mean MRE decreases from 1.76% to 0.23% for apparent resistivity, and from 0.56% to 0.40% for phase.

D. Computing Time and Resources Consumption

In this section, we intend to compare the time and memory consumption between SPI and FDM under comparable accuracy, which is controlled by the mesh scale (that is, the number of grids in both directions). For SPI, the uniform grid as introduced in this work is adopted, while for FDM, we

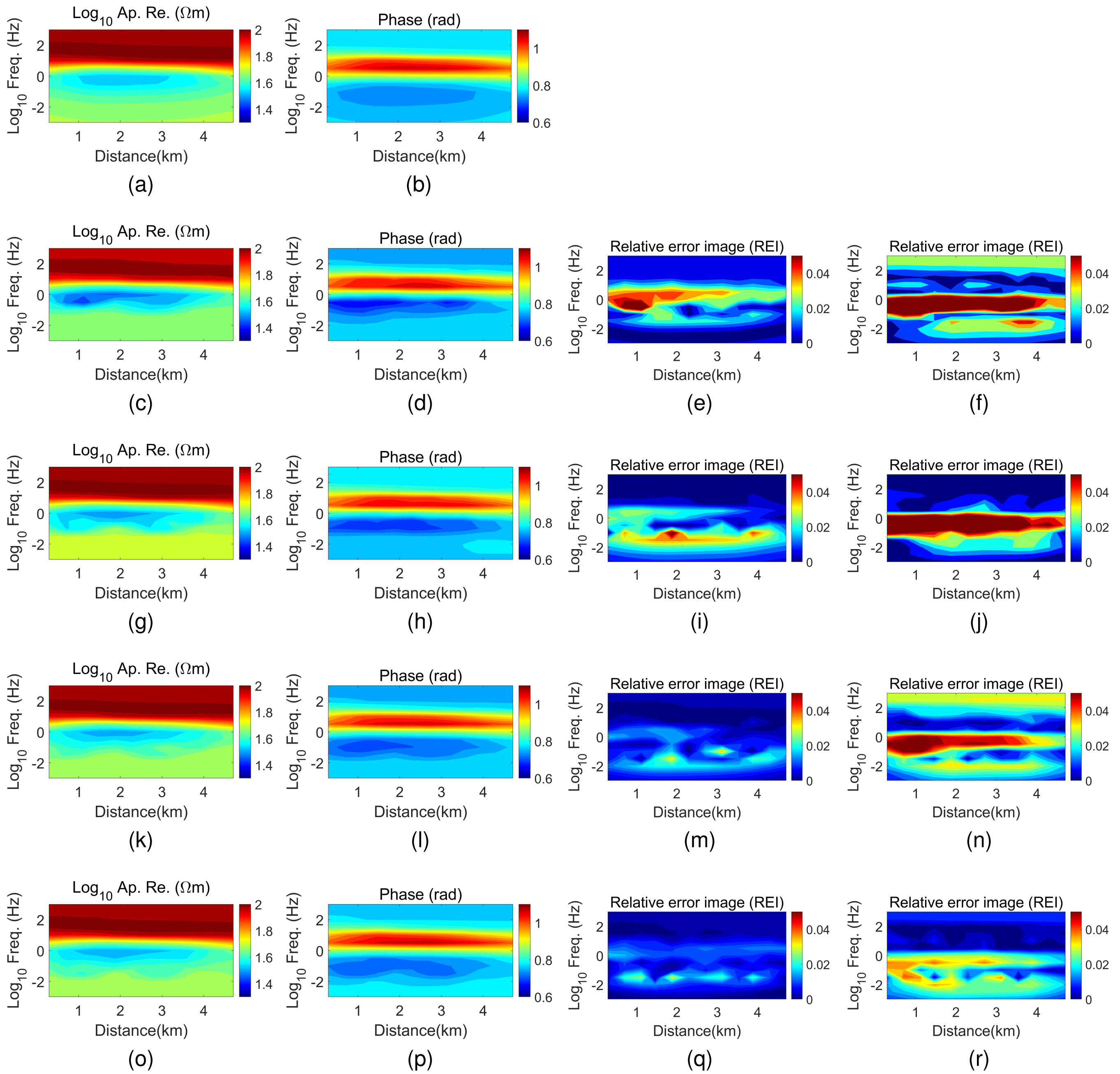

Fig. 6. (a) and (b) True logarithm apparent resistivity and phase sections of the Fig. 2 model approximated by the FDM (300, 500) solver. For the lower sixteen subfigures, the four columns from left to right respectively show apparent resistivity sections, phase sections, relative error images of apparent resistivity, and relative error images of phase, respectively corresponding to different SPI solver settings. Specifically, (c)–(f) correspond to SPI with $(N, N_{\text{path}}) = (30, 0.1\text{M})$, (g)–(j) correspond to SPI (40, 0.3M), (k)–(n) correspond to SPI (60, 0.5M), and (o)–(r) correspond to SPI (80, 1M).

use nonuniform grid since it can better handle multifrequency modeling compared to a uniform grid. We report the time and memory consumption of the FDM with three parameter combinations as introduced in Section III-C, and SPI methods with four parameter combinations: (N, N_{path}) = (30, 0.1M), (40, 0.3M), (60, 0.5M), and (80, 1M). The results are summarized in Table II. We used a Linux CPU server with two Intel[1] Xeon[1] Gold 5118 CPU@2.30 GHz and 512-GB RAM. In addition, performance comparisons are carried out using an NVIDIA A100 GPU. All the computations are performed on MATLAB R2021b. We report six time metrics and two memory metrics, include the following.

- MCT: Time consumption of constructing the path matrix using MC simulation on CPU (the first subprocess as introduced in Section II-C.1).
- MMT: Time consumption of constructing mapping matrices on CPU (the second subprocess introduced in Section II-C.2).
- LCT: Time consumption of loading the path matrix from disk to RAM.

[1]Registered trademark.

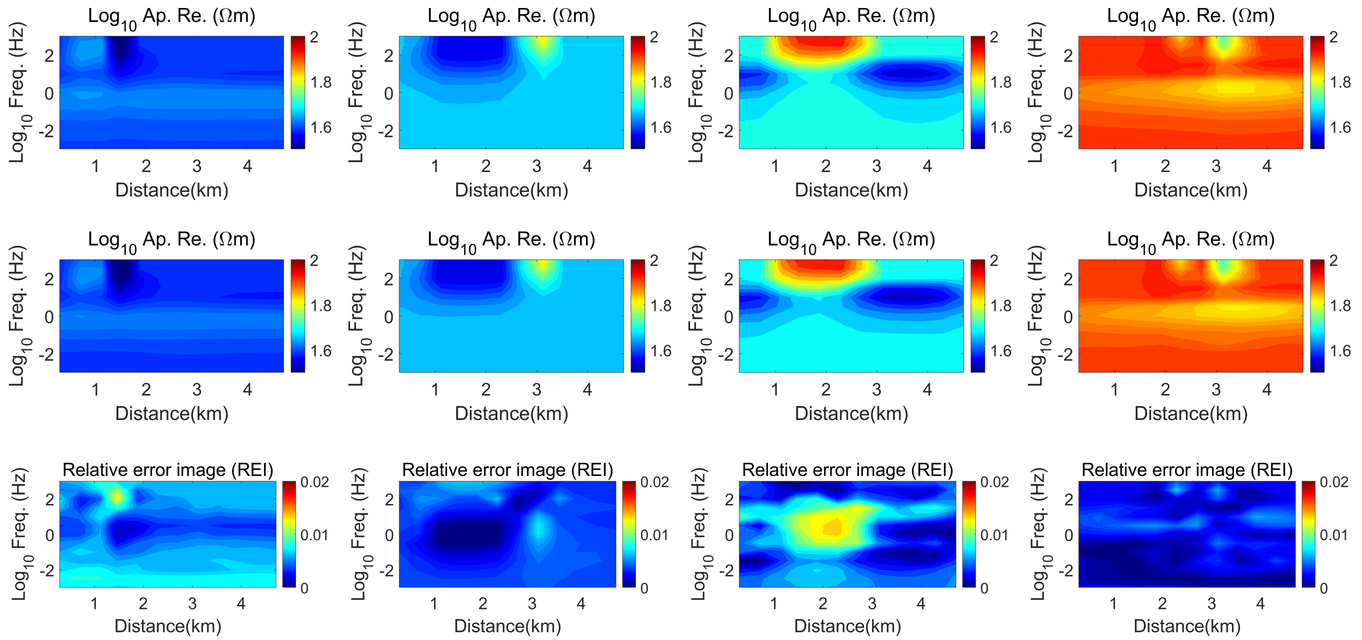

Fig. 7. Three rows from top to bottom represent the approximated true apparent resistivity, apparent resistivity computed SPI (N = 80, N_{path} = 1M), and relative error images. The four columns from left to right, respectively, correspond to the four conductivity models in Fig. 3.

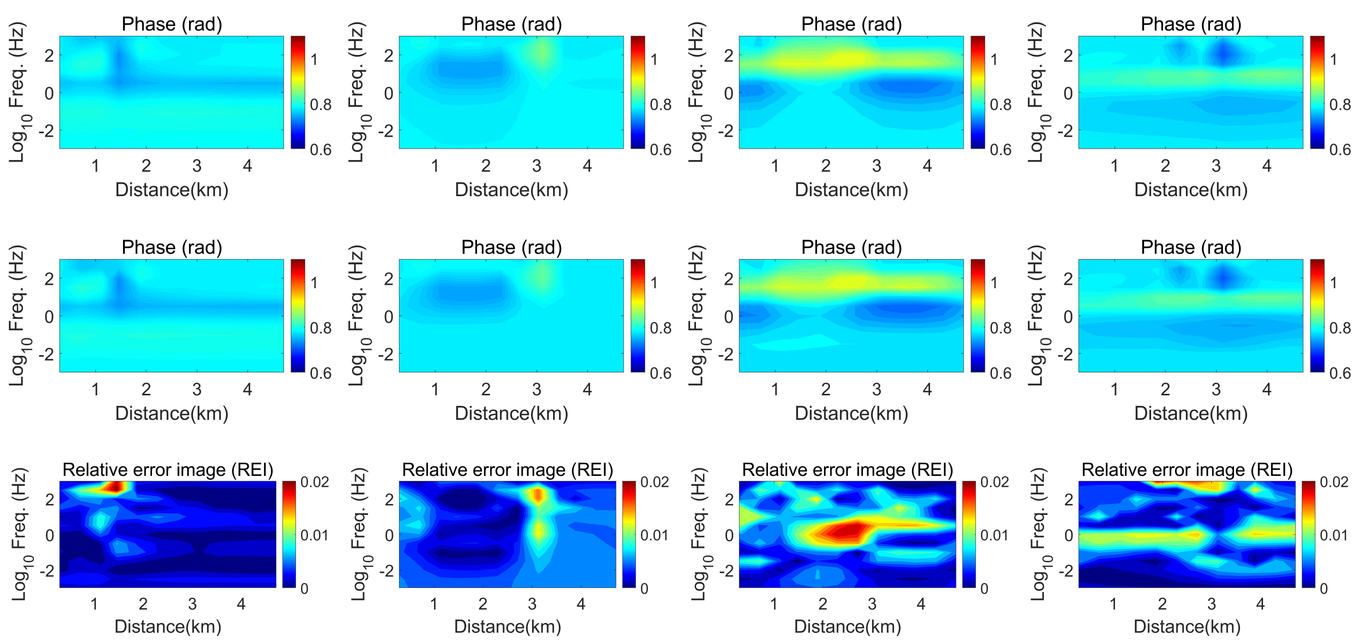

Fig. 8. Three rows from top to bottom represent the approximated true phase, phase computed by SPI (N = 80, N_{path} = 1M), and relative error images. The four columns from left to right, respectively, correspond to the four conductivity models in Fig. 3.

- LGT: Time consumption of loading the path matrix from RAM to GPU.
- CT: Online computation time on CPU (the third subprocess introduced in Section II-C.3).
- GT: Online computation time on GPU (the third subprocess introduced in Section II-C.3).
- AM: The average memory consumption during the online computation.
- PM: The peak memory consumption during the online computation.

The online computation of SPI (30, 0.1M), (40, 0.3M), and (60, 0.5M) is conducted on both the CPU server and the A100 GPU. Due to hardware limitations, we only tested SPI (80, 1M) on the CPU server. As shown in Table II, the preparation phase of SPI (including MCT, MMT, LCT, and LGT) is time-consuming, reaching several hundreds to thousands of seconds, depending on the SPI parameter size. Among these, MCT and LCT take up the most to the total preparation time. Once the required matrices are generated offline and loaded into RAM or GPU memory, the SPI can be conducted with lower

TABLE I

COMPARISON OF THE MREs FOR THE 2000 CASES BETWEEN FDM AND SPI SIMULATIONS. (APP: APPARENT RESISTIVITY, PHA: PHASE, FD: FULL DATA, AND ALL VALUES ARE IN UNITS OF 10^{-3}. SPI (N, N_{path}) MEANS THAT THE SPI COMPUTATION IS CONDUCTED WITH N_{path} RANDOM WALK PATHS, AND THE SHAPE OF THE MC MESH IS $(N+5)\times(N+1)$. FDM (N_{Fz}, N_{Fx}, d_{F}, k_{F}) MEANS THAT THE FDM MESH HAS N_{Fx} UNIFORM HORIZONTAL GRIDS AND N_{Fz} NONUNIFORM VERTICAL GRIDS, WHOSE THICKNESSES FORM A GEOMETRIC SEQUENCE WITH THE FIRST TERM OF d_{F} METERS AND A COMMON RATIO OF k_{F})

Method	Mean (app MRE)	Std. (app MRE)	Mean (pha MRE)	Std. (pha MRE)	Mean (fd MRE)	Std. (fd MRE)
SPI (30, 0.1 M)	13.7	2.9	10.0	4.7	11.9	2.8
SPI (40, 0.3 M)	6.1	3.3	9.5	4.3	7.8	3.4
SPI (60, 0.5 M)	4.6	2.6	7.1	3.7	5.8	3.0
SPI (80, 1.0 M)	5.5	3.0	6.6	3.0	6.0	2.7
FDM (25, 30, 19.3, 1.13)	17.6	4.3	5.6	3.8	11.6	3.8
FDM (50, 60, 10.3, 1.06)	8.7	2.7	2.8	2.4	5.8	2.4
FDM (100, 120, 4.93, 1.03)	2.3	3.2	4.0	3.3	3.2	3.2

TABLE II

COMPARISON OF THE TIME AND MEMORY CONSUMPTION BETWEEN FDM AND SPI. THE MEANINGS OF SPI (N, N_{path}), AND FDM (N_{Fz}, N_{Fx}, d_{F}, k_{F}) ARE THE SAME AS ABOVE

Method	MCT (s)	MMT (s)	LCT (s)	LGT (s)	CT (s)	GT (s)	AM (GB)	PM (GB)
SPI (30, 0.1 M)	271	8	7.46	1.65	2.30	0.032	7.5	7.6
SPI (40, 0.3 M)	625	11	33.88	8.60	9.85	0.16	21.5	29.5
SPI (60, 0.5 M)	1341	14	146	11.82	33.7	4.32	59.6	80.7
SPI (80, 1.0 M)	2108	17	671	/	104	/	180.0	270.0
FDM (25, 30, 19.3, 1.13)	/	/	/	/	0.12	17.78	10.8	11.4
FDM (50, 60, 10.3, 1.06)	/	/	/	/	0.16	20.41	10.8	12.0
FDM (100, 120, 4.93, 1.03)	/	/	/	/	0.44	27.24	10.8	15.0

time consumption compared with FDM, particularly when the accuracy requirement is modest. For example, FDM (25, 30, 19.3, 1.13) and SPI (30, 0.1M) yield comparable mean MRE levels of approximately 1.2%. The online computation time of SPI on GPU is 0.032 s, achieving over 550× speedup relative to GPU-based FDM, while its memory usage is only about 70% of that of FDM. SPI (60, 0.5M) attains a similar mean MRE to FDM (50, 60, 10.3, 1.06) of 0.58%, achieving about 5× speedup compared with GPU-based FDM and a larger memory usage. FDM (100, 120, 4.93, 1.03) achieves a mean MRE of 0.32% using less than 15-GB memory consumption, whereas SPI with comparable accuracy was not realized in our experiment due to resource limitations.

We observe from Table II that the speedup of SPI on GPU compared with CPU is substantial. This is primarily because the core computation of MT-SPI is matrix–vector multiplication, an operation that is highly optimized and efficiently executed on GPUs. In contrast, solving linear systems are less suited to GPUs compared with matrix–vector multiplication. For our experiments, we observe that the solution time of FDM on a single GPU is even longer than that on a CPU. Even with extensive engineering optimization, it is likely that the level of GPU acceleration achievable for solving linear systems remains lower than matrix–vector multiplication.

IV. DISCUSSION

In this work, we study the application of a stochastic algorithm for modeling diffusive EM fields in inhomogeneous media. We aim to investigate whether EM simulations can benefit from the combination of stochastic algorithms and parallel heterogeneous computing architectures. Our current answer is affirmative, as SPI achieves over 550× acceleration with less than 70% memory consumption on GPU under the MATLAB-based implementation, at the relative error level of 1.2%. However, because the convergence of SPI is considerably slower than FDM, SPI's demand for memory increases rapidly under high precision requirements, and the calculation speed will become less advantageous over or even slower than FDM.

A promising application scenario for SPI is Bayesian inversion, where tens of thousands of forward and gradient evaluations are typically required per inversion [37]. In this context, SPI is able to provide approximate results at a lower cost. In addition, SPI computes the field at a single observation point per run, whereas conventional methods compute the field across the entire computational domain. This makes SPI advantageous when only a few observation points are of interest.

The core value of SPI lies in its inherently parallel nature. The advantages of SPI over deterministic methods (e.g., FDM) can be illustrated from two perspectives. First, on a single GPU, the main computational cost in FDM comes from matrix decomposition, such as LU decomposition under a MATLAB-based implementation, whereas in SPI it comes from matrix–vector multiplication. Matrix–vector multiplication is one of the most computationally effective operations on GPUs, while sparse operations are not so well-optimized for GPU structures due to sequential operations and irregular memory access. In addition, since each random walk is fully independent, SPI scales almost linearly with the number of GPU devices, without suffering from the communication bottlenecks often encountered in traditional solvers. This implies that, in principle, given sufficient computing resources, SPI has the potential to achieve arbitrary fast runtimes even for a single simulation, with speedup approximately proportional to the number of computing units. The latter advantage is not evident in our experiment, as 2-D MT modeling is relatively small and does not require inter-GPU communication even for FDM. In

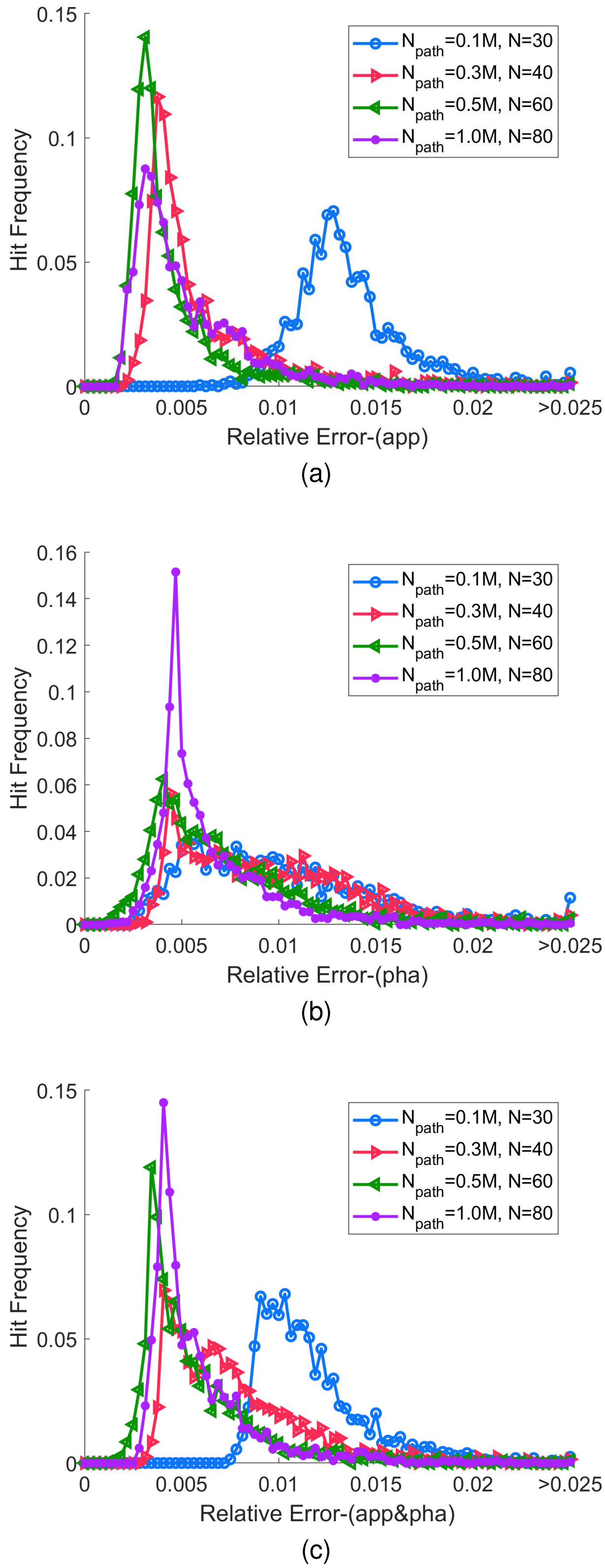

Fig. 9. Hit frequencyof the relative error between approximated true response and SPI ($N = 80$, $N_{\text{path}} = 1\text{M}$) simulations for 2000 models in the test model set. (a)–(c) Apparent resistivity (App), phase (Pha), and both apparent resistivity and phase (App&Pha), respectively.

the future, we plan to apply SPI and other stochastic methods to larger scale problems to further assess and demonstrate their potential.

We acknowledge that this work can still be further improved in several aspects. First, the implementation of the MT-SPI algorithm can be significantly optimized to enhance its performance. For instance, it may be feasible to reuse a single v for multiple nearby frequencies, thereby reducing the number of matrix–vector multiplications. It is also worth exploring whether a single nonuniform MC mesh can be used to model all frequencies consistently. Second, this work provides only a qualitative analysis of how SPI parameters affect computational accuracy. A more detailed and comprehensive sensitivity study could offer deeper insights into the algorithm's behavior and performance. Third, the hyperparameter α used in this study may not generalize across different scenarios. Its optimal value may depend on factors such as the average resistivity of the survey domain, the MC mesh dimension, and the frequency. Future research on the selection and generalization of α could contribute to the development of more robust SPI algorithms. This article marks a very beginning of our research on stochastic EM modeling. We will continue to conduct more in-depth research on this topic in the future.

V. Conclusion

In this work, the SPI is applied to 2-D TE mode MT modeling. The probabilistic solution to the Helmholtz equation is derived using the Feynman–Kac formula, transforming the task of solving the matrix equation into computing the expectation of the random walk path integral obtained by MC simulation. A two-stage scheme is developed, consisting of offline MC simulation, mapping matrices construction, and online matrix–vector multiplication. Models are mapped to the random walk mesh in different ways to handle multireceiver and multifrequency simulations. Numerical experiments demonstrate the accuracy and stability of SPI, verifying its high compatibility with GPU and the potential to reduce both the time and memory consumption.

Appendix
Introduction of the MT-FDM Solver

With the FDM solver, the subsurface conductivity model is discretized into a nonuniform mesh. The computational domain is divided into N_{Fx} grids in the horizontal direction and N_{Fz} grids in the vertical direction. To model the high-frequency attenuation, the mesh is extended in the depth direction with an additional 50 grids, where the thickness grows logarithmically, with the deepest grid having a thickness of 10 km. In the lateral directions, the mesh is extended uniformly by 20 grids in each direction. The air is modeled with ten grids in the upper direction. As a result, the total size of the FDM mesh is $(N_{\text{Fz}} + 60) \times (N_{\text{Fx}} + 40)$. The upper boundary, lower boundary, and side boundaries are, respectively, assigned a Dirichlet BC with a value of 1, a Dirichlet BC with a value of 0, and Neumann BCs with a value of 0.

For different frequencies, we solve a set of linear equations, that is,

$$\mathbf{A}(\omega)\mathbf{x} = \mathbf{b} \tag{25}$$

where ω is the angular frequency. We observe from (3) that all the $\mathbf{A}$s are square, symmetric, and share the same sparse structure. In addition, the nondiagonal elements of $\mathbf{A}$s are constant and independent of ω, while the diagonal elements Diag($\mathbf{A}$) vary linearly with ω

$$\mathrm{Diag}(\mathbf{A}) = \mathbf{a} - j\mu\omega\boldsymbol{\sigma} \tag{26}$$

where $\mathbf{a}$ is a constant vector related to the grid sizes and $\boldsymbol{\sigma}$ is the conductivity vector. Based on this property, in our implementation, the FDM computation is divided into two stages.

In the first stage, the vectors of row and column indices ($\mathbf{i}_s$ and $\mathbf{j}_s$) of the nonzero elements in $\mathbf{A}$s are generated, as well as the vector of nondiagonal elements in $\mathbf{A}$ and $\mathbf{a}$. These are all used to prepare for the second stage computation. In the computation of each model, this stage is conducted only once, and all the possible loop operations are replaced with matrix operations to maximize GPU computational efficiency.

In the second stage, the linear equations are solved in parallel using the `parfor` command, where the parallelization is performed over frequencies. For each frequency, $\mathbf{i}_s$, $\mathbf{j}_s$, and $\mathbf{a}$, nonzero elements and the conductivity vector are combined to assemble the stiffness matrix. The left division operator in MATLAB (\) is used to solve (25), which is carried out using a direct solver, like Cholesky or LU decomposition. The same code is used on both CPU and GPU, while for GPU computation, all quantities involved are transferred to the GPU through the `gpuArray` command.

REFERENCES

[1] M. N. Berdichevsky and V. I. Dmitriev, *Magnetotellurics in the Context of the Theory of Ill-Posed Problems*. Tulsa, OK, USA: Society of Exploration Geophysicists, 2002.

[2] R. L. Mackie, J. T. Smith, and T. R. Madden, "Three-dimensional electromagnetic modeling using finite difference equations: The magnetotelluric example," *Radio Sci.*, vol. 29, no. 4, pp. 923–935, Jul. 1994.

[3] Z. Xiong, "Electromagnetic modeling of 3-D structures by the method of system iteration using integral equations," *Geophysics*, vol. 57, no. 12, pp. 1556–1561, Dec. 1992.

[4] J. H. Coggon, "Electromagnetic and electrical modeling by the finite element method," *Geophysics*, vol. 36, no. 1, pp. 132–155, Feb. 1971.

[5] J. Xie et al., "3-D magnetotelluric inversion and application using the edge-based finite element with hexahedral mesh," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021.

[6] N. Bai, B. Han, X. Hu, J. Zhou, W. Liao, and G. Huang, "Efficient solution scheme for large-scale anisotropic forward modeling of 3-D magnetotelluric data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5905214.

[7] N. Yu, H. Zhang, H. Chen, W. Kong, X. Feng, and Y. Qian, "Three-dimensional unstructured finite element modeling of magnetotelluric problems allowing for continuous variation of conductivity in each block," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 2004412.

[8] G. R. Jiracek, R. P. Reddig, and R. K. Kojima, "Application of the Rayleigh-FFT technique to magnetotelluric modeling and correction," *Phys. Earth Planet. Interiors*, vol. 53, nos. 3–4, pp. 365–375, Mar. 1989.

[9] F. Jiang and D. Yang, "Fast 3D magnetotelluric modeling using a local mesh approach," in *Proc. SEG Tech. Program Expanded Abstr.*, Sep. 2020, pp. 585–589.

[10] T. Shan, R. Guo, M. Li, F. Yang, S. Xu, and L. Liang, "Application of multitask learning for 2-D modeling of magnetotelluric surveys: TE case," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–9, 2021.

[11] X. Wang, P. Jiang, F. Deng, S. Wang, R. Yang, and C. Yuan, "Three-dimensional magnetotelluric forward modeling through deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5916413.

[12] C. T. Wolfe, U. Navsariwala, and S. D. Gedney, "A parallel finite-element tearing and interconnecting algorithm for solution of the vector wave equation with PML absorbing medium," *IEEE Trans. Antennas Propag.*, vol. 48, no. 2, pp. 278–284, Feb. 2000.

[13] S.-C. Lee, M. N. Vouvakis, and J.-F. Lee, "A non-overlapping domain decomposition method with non-matching grids for modeling large finite antenna arrays," *J. Comput. Phys.*, vol. 203, no. 1, pp. 1–21, Feb. 2005.

[14] Z. Peng, X.-C. Wang, and J.-F. Lee, "Integral equation based domain decomposition method for solving electromagnetic wave scattering from non-penetrable objects," *IEEE Trans. Antennas Propag.*, vol. 59, no. 9, pp. 3328–3338, Sep. 2011.

[15] T. Cwik, R. Van De Geijn, and J. Patterson, "Application of massively parallel computation to integral equation models of electromagnetic scattering," *J. Opt. Soc. Amer. A, Opt. Image Sci.*, vol. 11, no. 4, pp. 1538–1545, 1994.

[16] Y. Zhang, M. Taylor, T. Sarkar, H. Moon, and M. Yuan, "Solving large complex problems using a higher-order basis: Parallel in-core and out-of-core integral-equation solvers," *IEEE Antennas Propag. Mag.*, vol. 50, no. 4, pp. 13–30, Aug. 2008.

[17] Q. Dinh and Y. Marechal, "Toward real-time finite-element simulation on GPU," *IEEE Trans. Magn.*, vol. 52, no. 3, pp. 1–4, Mar. 2016.

[18] C. H. Wu, "A multicolour SOR method for the finite-element method," *J. Comput. Appl. Math.*, vol. 30, no. 3, pp. 283–294, Jul. 1990.

[19] R. Tavakoli and P. Davami, "A new parallel Gauss–Seidel method based on alternating group explicit method and domain decomposition method," *Appl. Math. Comput.*, vol. 188, no. 1, pp. 713–719, May 2007.

[20] B. He, N. Chen, D. Lin, C. Lu, and P. Zhou, "An efficient parallel computing method for the steady-state analysis of low-frequency electromagnetics using an anti-periodic condition," *IEEE Trans. Magn.*, vol. 56, no. 3, pp. 1–4, Mar. 2020.

[21] C. Ding, Y. Zhou, W. Cai, X. Zeng, and C. Yan, "A path integral Monte Carlo (PIMC) method based on Feynman–Kac formula for electrical impedance tomography," *J. Comput. Phys.*, vol. 476, Mar. 2023, Art. no. 111862.

[22] H. Zhou, M. Li, F. Yang, S. Xu, and A. Abubakar, "Modeling magnetotelluric surveys based on stochastic path integral," in *Proc. IEEE Int. Symp. Antennas Propag. INC/USNC-URSI Radio Sci. Meeting (AP-S/INC-USNC-URSI)*, Jul. 2024, pp. 2545–2546.

[23] M. Kac, "On distributions of certain Wiener functionals," *Trans. Amer. Math. Soc.*, vol. 65, no. 1, pp. 1–13, Jan. 1949.

[24] M. I. Freidlin, *Functional Integration and Partial Differential Equations*. Princeton, NJ, USA: Princeton Univ. Press, 1985.

[25] R. Janaswamy, *Engineering Electrodynamics: A Collection of Theorems, Principles and Field Representations*. Bristol, U.K.: IOP Publishing, 2020.

[26] M. K. Chati, M. D. Grigoriu, S. S. Kulkarni, and S. Mukherjee, "Random walk method for the two- and three-dimensional laplace, Poisson and Helmholtz's equations," *Int. J. for Numer. Methods Eng.*, vol. 51, no. 10, pp. 1133–1156, Aug. 2001.

[27] R. Janaswamy, "Solution of BVPs in electrodynamics by stochastic methods," in *Proc. IEEE Appl. Electromagn. Conf. (AEMC)*, Dec. 2007, pp. 1–4.

[28] B. V. Budaev and D. B. Bogy, "A probabilistic approach to wave propagation and scattering," *Radio Sci.*, vol. 40, no. 6, pp. 1–11, Dec. 2005.

[29] M. S. Zhdanov, R. B. Smith, A. Gribenko, M. Cuma, and M. Green, "Three-dimensional inversion of large-scale EarthScope magnetotelluric data based on the integral equation method: Geoelectrical imaging of the Yellowstone conductive mantle plume," *Geophys. Res. Lett.*, vol. 38, no. 8, pp. 1–7, Apr. 2011.

[30] D. Yang, D. W. Oldenburg, and E. Haber, "3-D inversion of airborne electromagnetic data parallelized and accelerated by local mesh and adaptive soundings," *Geophys. J. Int.*, vol. 196, no. 3, pp. 1492–1507, Mar. 2014.

[31] Y. Wang, J. Liu, R. Liu, R. Guo, and D. Feng, "Frequency-domain magnetotelluric footprint analysis for 3D earths," *J. Geophys. Eng.*, vol. 16, no. 6, pp. 1151–1163, Dec. 2019.

[32] M. N. Nabighian, *Electromagnetic Methods in Applied Geophysics: Voume 1, Theory*. Tulsa, OK, USA: Society of Exploration Geophysicists, 1988.

[33] Y. Jin, Y. Ge, J. Wang, G. Heuvelink, and L. Wang, "Geographically weighted area-to-point regression Kriging for spatial downscaling in remote sensing," *Remote Sens.*, vol. 10, no. 4, p. 579, Apr. 2018.

[34] H. Zhou et al., "An intelligent MT data inversion method with seismic attribute enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4500714.

[35] H. Zhou et al., "Super-resolution magnetotelluric data inversion with seismic texture constraint," *Geophysics*, vol. 90, no. 3, pp. WA153–WA168, May 2025.

[36] U. M. Ascher and C. Greif, *A First Course on Numerical Methods*. Philadelphia, PA, USA: SIAM, 2011.

[37] X. Zhang and A. Curtis, "Seismic tomography using variational inference methods," *J. Geophys. Res., Solid Earth*, vol. 125, no. 4, Apr. 2020, Art. no. e2019JB018589.